# The General Divisor Theorem: Exact Density Correction under Residue Conditioning

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**Draft v3.3** — theorem-focused revision of Draft v3.2. The separate reading-point correspondence example has been moved to its own mathematical-basis note; the General Divisor Theorem and its proofs are unchanged.

### Abstract

Let $N, m$ be positive integers and $a$ an integer. Write

$$d = \operatorname{rad}(\gcd(m, N)), \qquad R = \frac{\operatorname{rad}(N)}{d}.$$

For

$$S(N, L, m, a) = \{1 \le n \le L : n \equiv a \pmod m,\ \gcd(n, N) = 1\},$$

we prove an exact dichotomy. If $\gcd(a, d) = 1$, the associated indicator has exact minimal period

$$T_{\min} = mR = \operatorname{lcm}(m, \operatorname{rad}(N)),$$

with exactly $\varphi(R)$ accepted values per period. Relative to the naive independent-density prediction

$$\frac{\varphi(N)}{N}\frac{L}{m},$$

the exact correction factor is

$$C(N, m) = \frac{d}{\varphi(d)},$$

independent of the admissible residue $a$. If $\gcd(a, d) > 1$, the accepted set is empty for every $L$. These period, count, and correction formulas are exact, rather than asymptotic, and require no divisibility or squarefreeness restriction on $N$ or $m$.

The theorem recovers the motivating primorial result when $N = P_k$ and $m \mid P_k$: squarefreeness gives $d = m$, $R = P_k/m$, $T_{\min} = P_k$, and $C(N, m) = m/\varphi(m)$.

### 1. Introduction / General Setup

Let $N$ and $m$ be positive integers and $a$ an integer. We study the counting problem

$$S(N, L, m, a) = \{1 \le n \le L : n \equiv a \pmod m,\ \gcd(n, N) = 1\},$$

the integers up to $L$ lying in a fixed residue class modulo $m$ and coprime to $N$.

A naive density heuristic treats the residue condition and the coprimality condition as independent: density $1/m$ from the residue class, multiplied by $\varphi(N)/N$ from coprimality to $N$. This gives the naive prediction

$$|S(N, L, m, a)| \approx \frac{\varphi(N)}{N}\frac{L}{m}.$$

This double-counts a restriction whenever $m$ and $N$ share a prime factor. If a prime $p$ divides both $m$ and $N$, and $n \equiv a \pmod{m}$ with $\gcd(a, p) = 1$, then $n$ is already guaranteed coprime to $p$ by the residue condition alone — the factor $\varphi(N)/N$ re-imposes that restriction a second time. The primes doing this double duty are exactly the primes shared between $m$ and $N$; primes of $N$ not dividing $m$, and primes of $m$ not dividing $N$, play no special role in this double-counting.

This motivates isolating the shared-prime structure directly, via

$$d = \operatorname{rad}(\gcd(m, N)), \qquad R = \frac{\operatorname{rad}(N)}{d},$$

the radical of the primes $m$ and $N$ have in common, and the radical formed from the primes of $N$ not shared with $m$. Section 2 states and proves the exact correction this produces. Section 3 walks through its specializations, recovering progressively more restrictive — and historically earlier — versions of the same result, down to the primorial case that originally motivated this work.

## 2. The General Divisor Theorem

### 2.1 A lemma used throughout

**Lemma** (exponent-blindness). For every integer $k \geq 1$,

$$\frac{k}{\varphi(k)} = \frac{\operatorname{rad}(k)}{\varphi(\operatorname{rad}(k))},$$

where $\operatorname{rad}(k)$ is the product of the distinct primes dividing $k$ ($\operatorname{rad}(1) = 1$).

**Proof.** For $k = 1$ both sides equal 1. For $k > 1$, write $k = \prod_i p_i^{e_i}$. For each prime power, $\varphi(p_i^{e_i})/p_i^{e_i} = 1 - 1/p_i$, independent of $e_i$. By multiplicativity,

$$\frac{k}{\varphi(k)} = \prod_i \frac{1}{1 - 1/p_i} = \prod_{p \mid k} \frac{p}{p-1},$$

which depends only on the set of primes dividing $k$ — i.e. only on $\operatorname{rad}(k)$. Applying the identical computation to $\operatorname{rad}(k)$ (all exponents 1) gives the same product. ■

### 2.2 Theorem 3 (General Divisor Theorem)

Let $N, m$ be any positive integers, $a$ any integer. Define

$$d = \operatorname{rad}(\gcd(m, N)), \qquad R = \frac{\operatorname{rad}(N)}{d}.$$

Let $S(L) = \{1 \leq n \leq L : n \equiv a \pmod{m},\ \gcd(n, N) = 1\}$.

**Case** $\gcd(a, d) = 1$**.** Writing $L = qT_{\min} + s$, $0 \leq s < T_{\min}$,

$$|S(L)| = q\,\varphi(R) + R_{\text{rem}}(s), \qquad T_{\min} = mR = \operatorname{lcm}(m, \operatorname{rad}(N)),$$

$T_{\min}$ is the exact minimal period of the indicator $\mathbb{1}[n \equiv a \pmod{m},\ \gcd(n, N) = 1]$, and against the naive predicted count $\frac{\varphi(N)}{N}\frac{L}{m}$,

$$C(N, m) = \frac{d}{\varphi(d)} = \prod_{p \mid \gcd(m, N)} \frac{p}{p-1}.$$

**Case** $\gcd(a,d) > 1$**.** Then $S(L) = \varnothing$ for every $L$.

**Proof.**

*Dichotomy and exact count.* Fix $n \equiv a \pmod m$. For each prime $p \mid d$: since $p \mid m$, $n \equiv a \pmod p$, so $n$ is coprime to $p$ iff $a$ is. If $\gcd(a,d) = 1$, every such $n$ is coprime to all of $d$; if some $p \mid d$ divides $a$, then $p$ divides every $n$ in the class, and since $p \mid d \mid N$, no such $n$ is coprime to $N$ — giving $S(L) = \varnothing$ for every $L$. Assume now $\gcd(a,d) = 1$: then $\gcd(n,N) = 1 \iff \gcd(n,R) = 1$ (the only primes of $N$ left to check are $R$'s). Since $\gcd(m,R) = 1$, any interval of $mR$ consecutive integers is a complete residue system mod $mR$; among such an interval, the sub-collection with $n \equiv a \pmod m$ numbers exactly $R$ elements and bijects with $\mathbb{Z}/R\mathbb{Z}$ via $n \bmod R$ (CRT), of which exactly $\varphi(R)$ are coprime to $R$. This holds for every such window, giving the exact count $\varphi(R)$ per length-$T_{\min}$ window and, summing over $q$ disjoint windows, $|S(qT_{\min})| = q\varphi(R)$.

*$T_{\min}$ is a period.* For $n \equiv a \pmod m$, $\gcd(n,N) = 1 \iff \gcd(n,R) = 1$ as above; for $n \not\equiv a \pmod m$, $n$ is unaccepted regardless. So the indicator factors as $I(n) = \nVdash[n \equiv a \pmod m] \cdot \nVdash[\gcd(n,R) = 1]$ for all $n$. The first factor has period $m$ (depends only on $n \bmod m$); the second has period $R$ (coprimality to $R$ depends only on $n \bmod R$, since $R$ is squarefree). Since $m \mid mR$ and $R \mid mR$, both factors, and hence their product, are unchanged under $n \mapsto n + mR$: $T_{\min} = mR$ is a period.

*No smaller $t$ is a period.* By CRT (using $\gcd(m,R) = 1$) pick $n_0 \equiv a \pmod m$, $n_0 \equiv 1 \pmod R$; $n_0$ is accepted. If $t > 0$ is any period, $n_0 + t$ is accepted, so $n_0 + t \equiv a \pmod m$, giving $m \mid t$. Fix a prime $p \mid R$ and suppose $p \nmid t$; by CRT choose $n_1 \equiv a \pmod m$, $n_1 \equiv -t \pmod p$, $n_1 \equiv 1 \pmod q$ for every other prime $q \mid R$. Then $n_1$ is accepted but $n_1 + t \equiv 0 \pmod p$ is not — contradicting periodicity. Hence $p \mid t$ for every prime $p \mid R$; since $R$ is squarefree, $R \mid t$, and with $m \mid t$, $\gcd(m,R) = 1$: $mR \mid t$. Thus $mR \mid t$ for every positive period $t$. Since $mR$ itself is a period, it is the exact minimal positive period: $T_{\min} = mR$.

*Ratio.* Write $N = AB$, $A = \prod_{p \mid N, p \mid m} p^{e_p}$ ($\mathrm{rad}(A) = d$), $B = \prod_{p \mid N, p \nmid m} p^{e_p}$ ($\mathrm{rad}(B) = R$), $\gcd(A,B) = 1$, so $\varphi(N) = \varphi(A)\varphi(B)$. By the Lemma (applied to $k = B$), $\varphi(B)/B = \varphi(R)/R$. Then

$$C = \frac{\varphi(R)/(mR)}{\varphi(N)/(Nm)} = \frac{N\varphi(R)}{R\varphi(N)} = \frac{AB\varphi(R)}{R\varphi(A)\varphi(B)} = \frac{AB\varphi(R)}{R\varphi(A)(B\varphi(R)/R)} = \frac{A}{\varphi(A)}.$$

By the Lemma once more (applied to $k = A$), $A/\varphi(A) = d/\varphi(d)$, giving $C = d/\varphi(d)$. ■

## 3. Specializations and Distinguishing Examples

Having proved Theorem 3 in full generality, we now walk down its specializations, each obtained by substituting a stronger hypothesis on $(N,m,a)$ directly into $d$, $R$, $T_{\min}$, and $C(N,m)$ — no independent re-derivation is required at any stage:

$$\text{Theorem 3} \longrightarrow \text{Theorem 2} \longrightarrow \text{Theorem 1}' \longrightarrow \text{Theorem 1},$$

where each arrow denotes "specializes to": the target theorem imposes stronger hypotheses, and its period, count, and correction-factor formulas are recovered from the source theorem by direct substitution, not by an independent proof.

### 3.1 Theorem 2 (specialization: $m \mid N$)

If $m \mid N$, then $\gcd(m,N) = m$, so $d = \mathrm{rad}(m)$ directly. Substituting into Theorem 3:

$$R = \frac{\mathrm{rad}(N)}{\mathrm{rad}(m)}, \qquad T_{\min} = m\,\frac{\mathrm{rad}(N)}{\mathrm{rad}(m)}, \qquad C(N,m) = \frac{\mathrm{rad}(m)}{\varphi(\mathrm{rad}(m))} = \frac{m}{\varphi(m)}$$

(the last equality by the Lemma applied to $k = m$), with admissibility $\gcd(a, \mathrm{rad}(m)) = 1$ and the same complementary zero branch. No new argument is required beyond this substitution; minimality, the exact count $\varphi(R)$, and the ratio are all inherited directly from Theorem 3.

### 3.2 Theorem $1'$ (specialization: $m$ unitary in $N$)

Call $m$ a *unitary divisor* of $N$ if $\gcd(m, N/m) = 1$ — equivalently, $m$ absorbs the full exponent of each of its primes in $N$. Under this hypothesis (a strengthening of $m \mid N$), write $N' = N/m$. Then $\mathrm{rad}(N) = \mathrm{rad}(m)\mathrm{rad}(N')$ (coprime factors), so $R = \mathrm{rad}(N)/\mathrm{rad}(m) = \mathrm{rad}(N')$ and

$$T_{\min} = m\mathrm{rad}(N'), \qquad C(N, m) = \frac{m}{\varphi(m)},$$

as in Theorem 2. The interval $N = mN'$ is also a period of the indicator (it is an integer multiple of $T_{\min}$, since $N'/\mathrm{rad}(N')$ is an integer), with total count $\varphi(N')$ over that longer interval — this matches Theorem 3's prediction exactly (using exponent-blindness to check $\varphi(N')/N' = \varphi(\mathrm{rad}(N'))/\mathrm{rad}(N')$), but $N$ **is generally not the minimal period**: it equals $T_{\min}$ only when $N'$ is itself squarefree.

### 3.3 Two distinguishing examples

$N = 18,\ m = 2$ — **coarser vs. minimal period.** Here $m = 2$ is a unitary divisor of $N = 18 = 2 \cdot 3^2$ ($\gcd(2, 9) = 1$), so Theorem $1'$ applies and gives a valid period of $N = 18$ with count $\varphi(9) = 6$. But $\mathrm{rad}(N') = \mathrm{rad}(9) = 3$, so Theorem 3's exact minimal period is

$$T_{\min} = m\mathrm{rad}(N') = 2 \cdot 3 = 6.$$

Direct enumeration confirms this: taking $a = 1$, $n \equiv 1 \pmod 2$ and $\gcd(n, 18) = 1$ (odd and not divisible by 3) gives $\{1, 5, 7, 11, 13, 17, \dots\}$ in $[1, 18]$, i.e. $\{1, 5, 7, 11, 13, 17\}$ — a pattern that visibly repeats every 6 integers, not every 18. So 18 is a genuine period (as Theorem $1'$ correctly states) but not the minimal one; 6 is.

$N = 5,\ m = 6,\ a = 2$ — **the sharp admissibility condition.** Here $\gcd(a, m) = \gcd(2, 6) = 2 \neq 1$, so this case would be *excluded* under Theorem 1's or Theorem $1'$'s hypothesis $\gcd(a, m) = 1$. But $d = \mathrm{rad}(\gcd(6, 5)) = \mathrm{rad}(1) = 1$, and $\gcd(a, d) = \gcd(2, 1) = 1$: admissible under Theorem 3's sharp condition. Here $R = \mathrm{rad}(5) = 5$ (since $5 \nmid 6$), $T_{\min} = mR = 30$. Direct check, $n \equiv 2 \pmod 6$ in $[1, 30]$: $\{2, 8, 14, 20, 26\}$; coprime to 5: $\{2, 8, 14, 26\}$ — count $4 = \varphi(5)$, matching the theorem. And $C(5, 6) = d/\varphi(d) = 1$, matching the naive prediction exactly ($\varphi(5)/5 \cdot (30/6) = 4$) — whereas $m/\varphi(m) = 6/\varphi(6) = 3$ would have been wrong. This confirms that $\gcd(a, d) = 1$ is the exact admissibility condition, not merely a cosmetic restatement of $\gcd(a, m) = 1$: it is strictly weaker in general, admitting strictly more values of $a$, since $d$ contains only the primes shared between $m$ and $N$.

### 3.4 Theorem 1 (specialization: primorial $N$)

Let $P_k = p_1 p_2 \cdots p_k$ denote the $k$-th primorial, the product of the first $k$ primes ($p_1 = 2, p_2 = 3, \dots$). By construction $P_k$ is squarefree. Let $m$ be a squarefree divisor of $P_k$ with $m > 1$, and let $a$ be an integer with $\gcd(a, m) = 1$.

If, in addition to $m \mid N$, $N$ is squarefree ($N = P_k$), then every divisor of $N$ — including $m$ — is automatically unitary, so Theorem $1'$ applies with $N' = P_k/m$; and since $P_k$ is squarefree, $N' = P_k/m$ is squarefree too (a divisor of a squarefree number is squarefree), so $\mathrm{rad}(N') = N'$ and the coarser and minimal periods of Section 3.2 coincide exactly:

$$T_{\min} = m\mathrm{rad}(N') = mN' = P_k.$$

Because $P_k$ is squarefree, $\mathrm{rad}(m) = m$, so $d = \mathrm{rad}(\gcd(m, P_k)) = \mathrm{rad}(m) = m$ exactly — the sharp admissibility condition $\gcd(a, d) = 1$ used in Theorem 3 is, in this primorial setting, *identical* to the hypothesis $\gcd(a, m) = 1$ above. This is the same pattern already seen in Theorem 2 ($d = \mathrm{rad}(m)$ whenever $m \mid N$); it collapses to an equality here specifically because $m$ is squarefree.

This gives:

**Theorem 1** (primorial specialization). Let $P_k$, $m$, and $a$ be as above. Write

$$L = qP_k + s, \qquad 0 \le s < P_k.$$

Then

$$|S(P_k, L, m, a)| = q\,\varphi(P_k/m) + R(s),$$

where

$$R(s) = \big|\{1 \le n \le s : n \equiv a \pmod m,\ \gcd(n, P_k) = 1\}\big|$$

is the exact count within the partial final period and satisfies $0 \le R(s) \le \lceil s/m \rceil$, with minimal period $P_k$ and correction factor

$$C(m) = \frac{m}{\varphi(m)},$$

independent of the admissible residue $a$. This is exactly Theorem 3 specialized as above, with no adjustment to period, count, or correction factor required.

**Independent primorial proof** The following is the original proof of Theorem 1, self-contained and independent of Theorem 3, retained here as the paper's historical starting point.

**Proof.** Since $m \mid P_k$ and $P_k$ is squarefree, $m$ and $P_k/m$ are coprime and squarefree, and $P_k/m$ consists exactly of the prime factors of $P_k$ that do not divide $m$.

Within one period $[1, P_k]$, exactly $P_k/m$ integers satisfy $n \equiv a \pmod m$. By the Chinese Remainder Theorem, because $m$ and $P_k/m$ are coprime, the residue of $n$ modulo $P_k/m$ ranges uniformly over all residue classes as $n$ ranges over its fixed class modulo $m$ within one period.

Hence exactly the fraction

$$\prod_{p \mid P_k/m} \Big(1 - \frac{1}{p}\Big) = \frac{\varphi(P_k/m)}{P_k/m}$$

of these $P_k/m$ values are additionally coprime to every prime dividing $P_k/m$. Since $\gcd(a, m) = 1$, they are already coprime to every prime dividing $m$. Therefore the exact count per period is

$$\frac{P_k}{m} \cdot \frac{\varphi(P_k/m)}{P_k/m} = \varphi(P_k/m).$$

The count does not depend on the specific value of $a$ beyond $\gcd(a, m) = 1$, because the argument uses only that the residue class modulo $m$ is fixed and coprime to $m$.

Using multiplicativity of Euler's totient function, $\varphi(P_k) = \varphi(m)\varphi(P_k/m)$, so $\varphi(P_k/m) = \varphi(P_k)/\varphi(m)$. The naive predicted count per period is $\varphi(P_k)/m$. The ratio is therefore

$$\frac{\varphi(P_k)/\varphi(m)}{\varphi(P_k)/m} = \frac{m}{\varphi(m)} = C(m).$$

This ratio is exact per full period and independent of $k$. Periodicity with period $P_k$ gives the general-$L$ statement with the bounded remainder above. ■

**Standardized refined-density identity** Throughout this paper, the refined predicted count is written in the equivalent forms

$$E_{\text{refined}}(L) = \frac{L}{m} \prod_{p \mid P_k/m} \Big(1 - \frac{1}{p}\Big) = L\,\frac{\varphi(P_k/m)}{P_k}.$$

These are equal because $\frac{L}{m} \cdot \frac{\varphi(P_k/m)}{P_k/m} = L\,\frac{\varphi(P_k/m)}{P_k}$.

**Corollary 1** ($m = 6$ case). For $m = 6$ and $a \in \{1, 5\}$,

$$C(6) = \frac{6}{\varphi(6)} = \frac{6}{2} = 3.$$

This corrects the earlier draft's reported limiting constant $24/25$ for the case $a = 5$.

Against the refined predicted count $E_{\text{refined}}(L) = L\varphi(P_k/6)/P_k$, the ratio is exactly 1.

*(In Section 2's general notation, $C(6) = 6/\varphi(6) = d/\varphi(d)$ with $d = 6$, since 6 is already squarefree — the two notations agree exactly, as they must by the remark above.)*

**Corollary 2** (Combined residue classes). Since the per-period count $\varphi(P_k/m)$ is independent of which admissible $a$ is used, the same ratio $C(m)$ holds for any union of admissible residue classes modulo $m$. For example, the classes $n \equiv 1$ or $5 \pmod 6$ together give the same ratio 3 as either class individually, because both the exact and naive counts scale by the same number of residue classes.

Both corollaries were verified numerically at $P_k = 210$ for $m \in \{6, 10, 30\}$, multiple values of $a$, and $L$ up to $10^7$, matching the predicted $C(m)$ to five decimal places or better in every case and matching the exact per-period counts $\varphi(P_k/m)$ precisely; see `tests/basis/test_persistent_constant.py`.

## 4. Numerical Verification

At $P_k = 210 = 2 \cdot 3 \cdot 5 \cdot 7$, $m = 6$, $a = 5$, the numerical results are:

**Table 1: Numerical verification at $P_k = 210$, $m = 6$, $a = 5$.**

| $L$ | actual | naive predicted | naive ratio | refined predicted | refined ratio |
|---|---|---|---|---|---|
| $10^5$ | 11,428 | 3,809.52 | 2.99985 | 11,428.57 | 0.99995 |
| $10^6$ | 114,285 | 38,095.24 | 2.99998 | 114,285.71 | 0.99999 |
| $10^7$ | 1,142,856 | 380,952.38 | 3.00000 | 1,142,857.14 | 1.00000 |
| $10^8$ | 11,428,571 | 3,809,523.81 | 3.00000 | 11,428,571.43 | 1.00000 |

Additional verification across $m \in \{6, 10, 30\}$ and all admissible $a$ mod $m$ confirms $C(m) = m/\varphi(m)$ exactly: $C(6) = 3$, $C(10) = 2.5$, $C(30) = 3.75$. It also confirms the exact per-period count $\varphi(P_k/m)$, independently of $a$, at $P_k = 210$ and $L$ up to $10^7$; see `tests/basis/test_persistent_constant.py`.

*(These values also follow directly from Section 2's general formulas: e.g. $C(10) = 10/\varphi(10) = 2.5$ and $C(30) = 30/\varphi(30) = 3.75$ reproduce exactly.)*

## 5. Conclusion

The General Divisor Theorem (Theorem 3) gives an exact, elementary correction to the naive independent-density heuristic for counting integers in a fixed residue class that are also coprime to a modulus — for arbitrary positive integers $N, m$ and any integer $a$, with no divisibility or squarefreeness restriction on either. Writing $d = \operatorname{rad}(\gcd(m, N))$ for the primes $m$ and $N$ share, and $R = \operatorname{rad}(N)/d$ for what remains, the indicator has exact minimal period $T_{\min} = \operatorname{lcm}(m, \operatorname{rad}(N))$, with exactly $\varphi(R)$ accepted values per period, whenever $\gcd(a, d) = 1$; the correction factor against the naive prediction is then exactly

$$C(N, m) = \frac{d}{\varphi(d)},$$

independent of which admissible $a$ is chosen. If instead $\gcd(a, d) > 1$, the accepted set is identically empty for every $L$ — a genuine dichotomy, not an edge case. The theorem follows from the Chinese Remainder Theorem and multiplicativity of $\varphi$, requiring no sieve-theoretic or analytic machinery, and its proof was independently audited and confirmed against exhaustive bounded computational tests, including the minimal-period claim; verification artifacts — along with the audit history — are available in the repository (Appendix A).

The primorial case that motivated this work is recovered exactly as one specialization: for $N = P_k$ squarefree and $m$ a squarefree divisor of $P_k$, squarefreeness forces $d = m$, giving

$$C(m) = \frac{m}{\varphi(m)}.$$

The primorial theorem is recovered without modification to its period, count, or correction factor. In its development, the $m = 6$ case also corrected an earlier reported constant $24/25$ to $C(6) = 3$; against the refined predicted count, the ratio is exactly 1.

## Appendix A. Verification Code

```python
from math import gcd
from sympy import totient, primefactors

def naive_predicted(limit, primorial, m):
    density = totient(primorial) / primorial
    return density * (limit / m)

def count_residue(limit, primorial, m, a):
    c = 0
    n = a
    while n <= 0:
        n += m
    for n in range(n, limit + 1, m):
        if gcd(n, primorial) == 1:
            c += 1
    return c

def refined_predicted(limit, primorial, m):
    # density = product over primes dividing Pk but NOT dividing m
    m_primes = set(primefactors(m))
    dens = 1.0
    for p in primefactors(primorial):
        if p not in m_primes:
            dens *= (1 - 1.0 / p)
    return dens * (limit / m)
```

The full runnable version of the primorial verification, including the general $(m, a)$ sweep, is in `tests/basis/test_persistent_constant.py`.

*(Added in v3.2.)* The corresponding exhaustive verification for the general theorem of Section 2 — covering arbitrary $(N, m, a)$, both branches of the dichotomy, arbitrary-window invariance, and divisor-based exact minimal-period detection over more than $4.7 \times 10^6$ checks, with zero counterexamples found — is in `test_general_divisor_theorem.py`, with results recorded in `verification-results.md`. The proof itself was independently audited line by line against a frozen specification (`theorem-specification.md`); a first pass (`proof-audit.md`) found two genuine gaps — an unproved arbitrary-window generalization and an unproved minimality premise — which were then closed and confirmed by a second, independent pass (`proof-audit-repaired.md`). Neither gap affected the primorial case (Section 3.4), which was independently reconfirmed as an exact specialization in `specialization-audit-repaired.md`.

## Appendix B. References


[1] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, Oxford University Press. Relevant background: the Chinese Remainder Theorem, Euler's totient function, and multiplicativity of $\varphi$.


## Appendix C. Project Vocabulary

The notation in this appendix is project-specific and is not part of Theorem 1. The theorem itself remains legible and valid independently of this vocabulary.

**Pi — expand.** $m \mapsto \mathbb{Z}/m\mathbb{Z}$. This exposes the complete residue/state space for a given modulus before any restriction is imposed.

$\pi$ — **extend.** $P_k \mapsto P_{k+1} = p_{k+1}P_k$. This adds another specification/constraint by extending the primorial by one further prime.

$\Pi$ — **resist.** $\prod_{p \mid P_k/m}(1 - 1/p)$. This is the fraction of candidates remaining admissible under the divisibility constraints not already imposed by the residue condition; it is the quantity appearing directly in the proof of Theorem 1.

These roles give the progression

$$\text{Pi (expand)} \to \pi \text{ (extend)} \to \Pi \text{ (resist)} \to r \text{ (read)}.$$

These names label roles already present in the proof of Theorem 1. The proof itself does not depend on this vocabulary and holds independently of it.

*(Added in v3.2.)* This vocabulary describes Theorem 1's proof specifically. The "extend" step ($P_k \to P_{k+1}$) does not have a natural analog in Theorem 3: the general theorem's $N$ has no canonical "next $N$" the way the primorial sequence does, so this vocabulary should not be read as describing the general theorem of Section 2.

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## Change Log (v3.2 → v3.3)

Draft v3.3 is a presentation-only revision. The General Divisor Theorem, its proof, specializations, numerical values, and verification claims are unchanged from Draft v3.2.

The former section "A Failed Correspondence as a Reading-Point Example" has been moved to a separate mathematical-basis note so that this paper remains focused on the divisor theorem. The abstract and conclusion were shortened accordingly, the numerical-verification and conclusion sections were renumbered, and Appendix A now points only to verification artifacts used by this paper.

No mathematical claim was changed in this revision.